

Inferno Makepad Rebuild Spec

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Summary

Rebuild the Inferno client in Rust with Makepad. Follow the Rails app for the look and for most features, follow Flutter for sync and protocol behavior, and fix the bugs both share instead of porting them.

The Nostr relays, LiveKit and Blossom stay as they are. Both sources come from the `flutter-rewrite` branch, which holds the Rails app at the root and Flutter in `lib/`. `main` is about four months behind it.

- **Rails** has the stronger visual system, and more of its features work end to end: group DMs, DM calls, notifications, a status picker, batch moderation, the onboarding wizard, rich search filters and a mobile drawer layout.
- **Flutter** is ahead on cross-device sync (kind 30078 config, NIP-65), Saved Messages, the key-backup flow, per-device voice state and relay-layer fixes, and it already ships on mobile.
- Features were judged by reading the code; neither app was run.

Target stack

Every piece is MIT, Apache-2.0 or BSD, so nothing needs attribution or a paid license. Because a Rust UI has no webview, the Linux WebRTC problem goes away.

Layer	Choice	Replaces (Flutter)	License
UI	Makepad 2.0 (Splash DSL), pinned to one <code>dev</code> commit	Flutter widgets, Riverpod	MIT / Apache-2.0
Nostr	<code>rust-nostr</code> (<code>nostr</code> , <code>nostr-sdk</code>), hand-rolled NIP-29 and custom kinds where missing	<code>lib/nostr</code> , <code>lib/crypto</code>	MIT
Voice / video	LiveKit Rust SDK (bundles <code>libwebrtc</code>)	<code>livekit_client</code>	Apache-2.0 / BSD

Layer	Choice	Replaces (Flutter)	License
Noise suppression	DeepFilterNet (native Rust, no FFI)	FFI bridge	MIT / Apache-2.0
NSFW model	<code>ort</code> (ONNX Runtime)	C bridge	MIT / Apache-2.0
Local cache	SQLite via rusqlite or sqlx	Drift	MIT
Key storage	<code>keyring</code> crate (desktop), platform keychain on mobile	<code>flutter_secure_storage</code>	MIT / Apache-2.0

Open items

- Makepad 2.0 is unreleased and moves fast. Pin the version and update on purpose.
- The remediation plan lists "AGPL vs ELv2" for Inferno's own license as unresolved (#111). That is a separate decision from the dependency licenses.

Visual spec: themes, surfaces, type

Use the Rails tokens, not Flutter's: Flutter reuses the theme names with different values. All seven themes are dark. In Makepad, put them in one theme object that shaders read live, so switching repaints the next frame.

Theme tokens (from `app/assets/stylesheets/application.tailwind.css`, lines 3 to 260)

Token	inferno	frostfire	boron	brimstone	plasma	pulsar	obsidian
gray-950	#0a0a09	#080f1a	#070f0a	#0d0814	#0e0710	#110c12	#0a0f1a
gray-900	#141312	#0f1b2d	#0f1f15	#1a1025	#1c0f1c	#1e141e	#111827
gray-800	#1e1c1b	#1a2a3a	#1a261c	#241a38	#2b1828	#2c1e2a	#1a2332
gray-700	#2c2a29	#243b53	#253328	#352750	#3d2438	#3e2a3c	#1e293b
gray-600	#403e3c	#3d5a80	#3a5040	#4d3a6a	#5c3a52	#5a3e54	#334155
gray-500	#656361	#627d98	#577057	#705a8a	#7d5570	#806078	#475569
gray-400	#878583	#829ab1	#748c74	#9078ab	#9b7089	#a07d94	#64748b
gray-300	#a8a7a5	#9fb3c8	#93ad93	#ad93c8	#b8899e	#c09cb2	#94a3b8
gray-200	#cccbca	#bcccdc	#b8cfb8	#ccb8e0	#dbb4d0	#e0c4d4	#cbd5e1
gray-100	#e1e0df	#d9e2ec	#d6e4d6	#e4d9f0	#f0d4e8	#f2dfe9	#e2e8f0

Token	inferno	frostfire	boron	brimstone	plasma	pulsar	obsidian
accent	#dc2626	#3b82f6	#10b981	#a855f7	#ec4899	#f9a8d4	#94a3b8
accent-light	#f87171	#93c5fd	#6ee7b7	#d8b4fe	#f9a8d4	#fccfe8	#cbd5e1
accent-dark	#b91c1c	#1d4ed8	#059669	#7e22ce	#db2777	#f472b6	#64748b
confirm	#ea580c	#2563eb	#059669	#7c3aed	#db2777	#f472b6	#3b82f6
danger	#dc2626	#ef4444	#ef4444	#ef4444	#ef4444	#ef4444	#ef4444

Some colors are the same in every theme:

- **Success:** #16a34a, light #4ade80.
- **Warning:** #eab308, light #facc15.
- **Presence:** online #22c55e with a 5px glow, idle = warning, dnd #ef4444, offline = gray-500.
- **Settings overlay:** gray-900 at 0.92 alpha with a 6px backdrop blur.
- **Obsidian:** turns off the pulse, flare and glow animations, and softens the rail glow to 4px at 0.2.

Surface map

Surface	Token
Server rail, user panel	gray-950
Window background, menus, voice bar, settings	gray-900
Channel sidebar, member list, modals, profile cards, toolbars	gray-800
Main chat column (one step lighter than the sidebars)	gray-700
Active channel, active settings item, composer bar	gray-600
Text: body / message / secondary / timestamp	gray-100 / gray-200 / gray-400 / gray-500

Typography (the system UI font; code in Consolas, Monaco, monospace)

Element	Size / weight / color
Message body	14px, line-height 1.625, gray-200
Emoji-only message	56px
Username	14px medium, top role color, underline on hover
Timestamp	12px gray-500; 10px on a grouped row's hover
Channel name	14px medium; unread = bold white
Category header	12px semibold, uppercase, wide tracking, gray-400
Server name, channel title	16px semibold white
Mention	accent at 15% background, accent text, radius 4, weight 500; @everyone and @here use warning colors
Code block	rgb(18,17,16) background, 1px gray-700 border, radius 8, padding 12, Monokai

Flip a theme with a 0.4s ease-in-out crossfade. The theme picker makes the settings overlay transparent so the live app previews behind it.

Visual spec: layout and sizing

The desktop shell is a row of five columns: a 72px rail, a 240px channel sidebar, a flexible chat column, and a 240px member list or 420px search panel on the right. Every header is 48px tall.

Region	Size	Details
Server rail	72px wide, 12px vertical padding, 8px gaps	Icons 48×48. Radius 16 at rest and 12 on hover or active, animated over 200ms. Hover fill is a gradient from accent-dark to accent. Hover glow: <code>0 8px accent/.4, 0 0 2px accent/.6</code> . Right edge: 1px <code>accent/.15</code> with a 12px glow at <code>accent/.06</code> .

Region	Size	Details
Rail indicators	Active pill 3×40px; unread pill 3×8px	Active pill is accent-light at left -10px and pulses over 2s ("ember pulse"). On hover it widens to 5px with a flare. Mention badge: min 18×18, 12px bold, 2px gray-950 border, glow 8px accent-medium/.5. Separator: 32×2px gray-800.
Server folders	48px tile	2×2 grid of 20px mini-icons, 10px label below. Expanding animates over 0.3s ease-out.
Channel sidebar	240px, gray-800, 8px padding, 2px between items	Header 48px with a gradient "lava seam" bottom border. Item padding 8×6, radius 4. Active item: 135° gradient (accent/.18 → accent-dark/.08), 2px accent left border, inset glow. Hover: accent/.08. Unread: 4×8px gray-300 pill.
Categories, voice	Header padding: 16 top, 4 bottom, 8 sides	Chevron rotates over 200ms. Voice participants: 20px avatars, indented 24px. Nested voice: indented 26px with L-shaped connectors that glow on 11s, 17s, 23s and 19s cycles so they never sync.
User panel	gray-950, 8px padding	32px avatar, 14px status dot (click opens the status picker). Name 14px medium, status 12px gray-400, version 10px gray-600.
Voice bar	gray-900	Sits above the user panel. Top border pulses from accent/.2 to accent-light/.4 over 3s. 20px icon buttons.
Chat header	48px, 16px side padding	Icon, name, a 1×24px divider, then the topic. Bottom border accent/.12. Pin button has a 16px accent badge.

Region	Size	Details
Member list	240px, gray-800, padding 16×8	Left border 1px accent/.15 with a soft glow. Item: 32px avatar, 12px status dot. Offline members at 40% opacity. Hover adds a gradient and a 2px accent/.5 left border. Role headers: 12px semibold uppercase in the role color, "Role — N".
Search panel	420px, gray-800	Slides over from the right.
Settings overlay	Full screen	224px gray-800 nav. Content max 768px, padding 32×40. Close button is a 36px circle with a 2px gray-600 border. Escape closes it.

On desktop between 768 and 1099px wide, Rails zooms everything to 0.85, and to 0.75 below 940px. In Makepad, use a UI scale factor instead.

Visual spec: messages and composer

A message row is a 40px avatar column plus a content column. Consecutive messages are grouped when four things hold: same author, neither is a system message, neither is a reply, and they are under 300 seconds apart. A grouped row replaces the avatar with a 40px spacer that shows the time on hover.

Message row (`messages/_message.html.erb`)

- **Row:** padding 2×8, radius 4. The list has 16px padding and 2px gaps.
- **Hover:** accent/.06 fill, a 2px accent/.4 left border, an inset glow, and an 8px accent/.35 glow on the avatar. Every transition takes 0.15s.
- **Avatar:** a 40px circle with a 16px right margin. With no image it shows the initial on the user's profile color (default #1e1c1b).
- **Reply preview:** a 12px gray-400 line above the name with the parent author in role color and the body cut at 60 characters. Clicking it jumps to the parent and flashes it at accent/.3, fading over 4s.
- **Hover toolbar:** floats at the top right of the row, half above it. It's gray-800 with radius 4, a 1px accent/.25 border, and the shadow `0 2px 12px rgba(0,0,0,.4), 0 0 8px accent/.1`. Actions: react, reply, pin (needs `manage_messages`), spoiler, hide, and edit

(own messages only).

- **Reactions:** pills with radius 6, padding 10×4, a 1px border, and 6px gaps. The emoji is 18px and the count gray-400. Your own reaction gets an accent/.15 fill and accent border; others are gray-800 with a gray-700 border.
- **Media:** images max 384×288 with radius 8, a shimmer while loading, and a 16px accent/.2 glow on hover. Video max 512×384. File and audio cards are gray-800 with radius 8.
- **Small bits:** a pinned label at 11px in warning-light/.7, "(edited)" at 12px gray-500, and new messages slide in 12px over 0.3s.

Composer (`channels/show.html.erb` , lines 145 to 235)

- **Bar:** gray-600, radius 8, with a permanent 1px accent/.2 border and 16px outer padding.
- **Focus glow:** `0 0 0 2px accent/.12, 0 0 20px accent/.1, inset 0 0 12px accent/.04`.
- **Text:** 14px with a 24px line height. The box grows to 192px, then scrolls. 20px icons; the send button glows on hover.
- **Bars above it:** reply, edit and spoiler bars sit on top. They're gray-700 with rounded top corners and an accent gradient on the left edge.
- **Typing indicator:** a 24px row of 12px gray-400 text with three dots blinking at 1.4s, staggered by 0.2s.
- **"New messages" bar:** a full-width confirm gradient that slides down over 0.3s.
- **Picker:** 384px wide and min(420px, 60vh) tall, with tabs for GIFs, Stickers and Emoji. The active tab has a 2px accent underline. On phones it becomes a bottom sheet.
- **Fire and electric effects:** keep Flutter's composer-edge effects as an option on top of the Rails composer. They're the one visual Flutter added that Rails lacks.

Visual spec: signature touches and mobile

What makes it feel like Rails is one faint accent glow repeated everywhere, plus short eased motion. Keep both, and keep them consistent.

Element	Spec
Scrollbars	3px and invisible at rest. Scrolling shows accent/.25, which fades to accent/.08 after 800ms idle and clears after another 600ms.

Element	Spec
Tooltips	gray-950, 12px semibold gray-100, padding 5×10, radius 5, an 8px arrow. 400ms delay, 100ms fade. Placed top first, then bottom, left, right, with a 6px gap.
Toasts	Top right with a 16px inset; success or danger fill, radius 8. They slide 80px in over 0.4s <code>cubic-bezier(0.16, 1, 0.3, 1)</code> and fade out after 4s.
Modals	Backdrop black/60 fading in over 0.2s. Dialog gray-800, radius 12, max 448px wide, rising from scale 0.92 and 16px down over 0.3s with the same easing.
Menus, dropdowns	gray-900, 1px gray-700 border, radius 8, padding 6, min 180px wide. Items 14px gray-300 that turn gray-700 and white on hover. Open from scale 0.9 over 0.2s.
Form controls	18px checkboxes with radius 6 and accent fill. Sliders: 6px track at accent/.2, 14px thumb. Text selection accent/.3, focus ring accent/.6.
Profile card	288px wide, 80px banner, 66px avatar with a 5px ring overlapping the banner by 35px. Role chips are black/30 with an 8px color dot. Full profile modal: 120px banner, 90px avatar.
Buttons	Confirm and danger buttons glow 14px accent/.5 on hover. Primary buttons use a confirm-dark → confirm gradient.
Spoilers, NSFW	<code>blur(60px) saturate(0)</code> , cleared on click over 0.3s. Inline spoilers are a solid #1e1e1e block.
Voice tiles	4:3, radius 12, a radial gradient in the user's color. While speaking, the ring and glow scale with audio level.
Drag and drop	A 2px accent drop line that appears over 150ms.
Tutorial hints	Pulsing accent arrows on a 2s cycle and a 260px tip card.

Mobile (Rails `mobile_nav_controller.js` ; Flutter has no equivalent)

- **Below 768px:** the rail and channel sidebar are hidden. A hamburger in the header slides both in together from the left over 0.25s, with the sidebar offset by the rail's 72px.
- **Below 1024px:** a header button slides the member list in from the right.
- **Closing drawers:** a black/50–60 backdrop closes them on tap, and switching channels closes them automatically.
- **Below 640px:** message rows drop the avatar column. A 28px avatar sits inline with the name and a short time, and side padding drops to 4px.
- **Hover-only controls:** the picker becomes a full-width bottom sheet, and controls that need hover show at 0.7 opacity on touch devices.
- **Makepad:** use the same three breakpoints and the same drawer motion.

Feature matrix

Of the 24 areas below, Rails is the reference for 15 and Flutter for 4. Either works for 2, DMs merge both, and 2 need new work. Status comes from reading the code of each app on `flutter-rewrite`.

Area	Rails	Flutter	Follow
Identity, keys	Working: setup, nsec import, NIP-49 export, NIP-05	Working, plus a backup step at signup, saved accounts, key-backup password	Flutter
Cross-device sync	Not present	Working: server list and settings in kind 30078, NIP-65 relay list, kind 0 merge	Flutter
Relay layer	Working	Working, with fixes for dedup, subscription leaks, stale copies reverting state	Flutter
Servers	Working: wizard, invites, discover, folders	Working, except folders (none) and delete (local only)	Rails, plus a real delete event

Area	Rails	Flutter	Follow
Server onboarding	Working: rules, roles, channels on join	Configured but never shown to joiners	Rails
Channels	Working: text, voice, encrypted, NSFW, post-only and verification channels, nesting, sidechat, bridged channels	Working, except post-only and verification; encrypted creation broken	Rails
Messaging	Working: markdown, edit, reply, pin, reactions, spoilers, mentions, jump-to	Working, same set	Rails for visuals, either for logic
Search	Working: from, in, has, before, after, pinned; people search	Partial: from and in only, server channels only	Rails
DMs	Working: NIP-44 kind 14, requests, pins, search	Working: kind 14, typing, encrypted reactions, Saved Messages; DM edit, delete and pin dead	Merge both, then add NIP-17
Group DMs	Working: create, add and remove members	Stub	Rails
Voice channels	Working: screen share, camera, DeepFilterNet, device select, server mute and move, request-to-speak	Partial: voice and devices work; video never rendered; push-to-talk not applied	Rails
Voice presence	Working	Working, per device with expiry	Flutter
DM calls	Working: ring, accept, decline	Missing	Rails
Roles	Working: multi-role, hoist, editor	Working	Either

Area	Rails	Flutter	Follow
Moderation	Working: kick, ban, timeout, batch actions, prune, history	Partial: kick, ban, prune; timeout and ban not enforced; no unban; no batch	Rails
Nicknames	Working	TODO	Rails
Presence, status	Working: online, idle, dnd, invisible, custom status with emoji, 15 minutes to auto-idle	Partial: auto-idle only, no picker	Rails
Friends, blocks	Working	Working, except block (never called)	Rails
Media, Blossom	Working, with a user-editable Blossom server list	Working, with servers hardcoded	Rails
Emoji, stickers, GIFs	Working, with emoji and stickers shared with members	Picker works; custom emoji never published	Rails
Notifications	Working: desktop alerts per preference, badges	Stub	Rails
Content safety	Working: Fire Shield, NSFW, hash lists, reputation, report wizard	Working: same pipeline, model loads off the main thread	Either; the report needs a real send, not clipboard
Keybinds	Stub	Stub	New work
Distribution	Electron plus Ruby sidecar, Tebako; no mobile	Native on all five, GitHub auto-update, store updater	New: Makepad packaging

Rails' admin panel was removed in commit 808cd74 and still has two dead links; don't port it.

Fix, don't port

These are wrong in at least one app and must be done correctly in the Rust core from day one. Most are in the protocol layer, so they also affect anyone still on the Flutter build.

Problem	Where today	Rust core rule
New encrypted channels send plaintext	Flutter: the create path never generates <code>channelPublicKey</code> (<code>server_settings_overlay.dart</code> 1514–1532)	Creating an encrypted channel generates and distributes the key before the first send; refuse to send if the key is missing
Custom emoji and stickers invisible to others	Flutter never publishes kinds 31754 and 31755	Publish on every add or delete
Bans and timeouts not enforced	Flutter publishes <code>timed_out_until</code> and bans but never applies them on receive	Enforce on receive: hide a banned user's messages, block sends during a timeout
Blocking doesn't filter messages	Flutter, remediation item #103	Blocked users are filtered in the core, not the UI
DMs not gift-wrapped	Both apps sign kind 14 directly; kind 1059 is subscribed but unhandled	Move to NIP-17: kind 14 sealed in 13, wrapped in 1059; read the old kind 14 during a migration window
Server delete is local only	Flutter	Publish a deletion so other members drop it
Video never displayed	Flutter has no video renderer	Render LiveKit frames to a Makepad texture
Unread state local only	Flutter seeds new channels as read (#62)	Sync read markers in the kind 30078 config

Problem	Where today	Rust core rule
Authority report is clipboard-only	Both (#109)	Out of scope for the first milestone; track separately

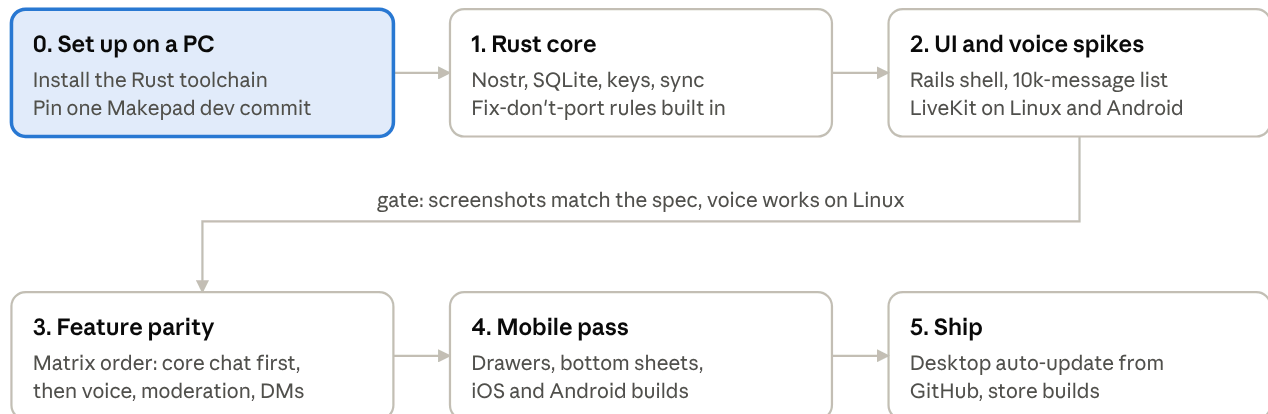
Custom kinds to keep stable (both apps already speak these, so they stay wire-compatible):

- **Server state:** 31750 metadata, 31751 structure, 31752 roles, 31753 members, 31754 emoji, 31755 stickers, 31756 bans, 31757 invites.
- **Live signals:** 10070 voice state, 25050 typing, 9006 pins (custom reuse), 30315 presence.
- **d-tag prefixes:** `inferno-` , `inferno-struct-` , `inferno-roles-` , `inferno-mbr-` .

Build order

Set up the toolchain on a PC first. After it, the two spikes decide whether Makepad holds up before any feature work starts.

Set up, prove the UI and voice, then reach parity



build order · 6 steps, 1 gate

If either spike fails the gate, the Rust core still carries over unchanged: it can sit under Flutter through `flutter_rust_bridge` or under a Tauri webview instead.

- ☐ Install the Rust toolchain on the PC
- ☐ Decide whether to merge `flutter-rewrite` into `main` (the auto-updater and README point at `main`)

- ☐ Pick the Makepad commit to pin
- ☐ Start the Rust core crate